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Submitted to ZARTHI assignment on Cloud Computing

Architecture Overview:

I used s3(Simple Storage Service) with AWS(Amazon Web Services) to host my Static Website.

AWS(Amazon Web Services): A cloud platform that offers service like storage, computing and databases over the internet.

S3(Simple Storage Service): Used to store and retrieve any amount of data and hosting Static Website.

Static Website Files:

Index.html: Landing or default web page, main page of the website that loads automatically.

Me.jpeg: An image of myself on the home page.

About.html: Consist of what I learned from this activity and difficulties I faced.

Style.css: Controls the look and style of the web pages.

Contact_us.html: A simple form where anyone can find a contact details of the creator.

In eager to practice more, I created Contact us page as well to make a webpage interactive.

➤ **While hosting a Static Website using s3 on AWS, I follow some process that involves:**

- Creating an account on Amazon free tier, and filled up some details for verification to access console of AWS(Amazon Work Services).
- There I access s3, an Object storage service and create a bucket named **“ankitkumarjha-portfolio-s3”**.
- I uploaded there some of my static website files like: index.html, about.html, contact_us.html, style.css, me.jpeg.
- After uploading all those files we need to enable **‘Static Website hosting’** inside my bucket to serve those files as a website, not just as stored files.
- Disabling **Block Public Access(bucket setting)** to make the content accessible for public but still I need to set who can do what in my bucket.
- Setting up the **bucket policy**, written in JSON, provides an access to the objects stored in the bucket. It allow public to read or write in our bucket and now our link is accessible.
- Enabling an extra feature of **‘Server Access Logging’** which help us to trace every request made to my bucket like who accessed it, what they did and when by specifying a new destination bucket so log don’t get messed up or cause a loop.
- **Enable versioning** help us to keeps multiple copies of a file, so if I change or delete a file by mistake, we can recover an older version.

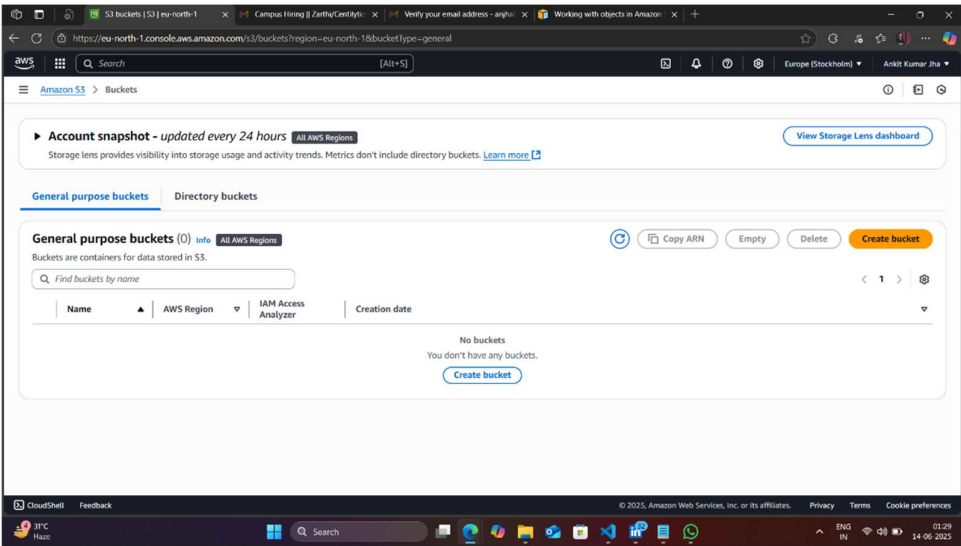
Working public URL:

<http://ankitkumarjha-portfolio-s3.s3-website.eu-north-1.amazonaws.com/>

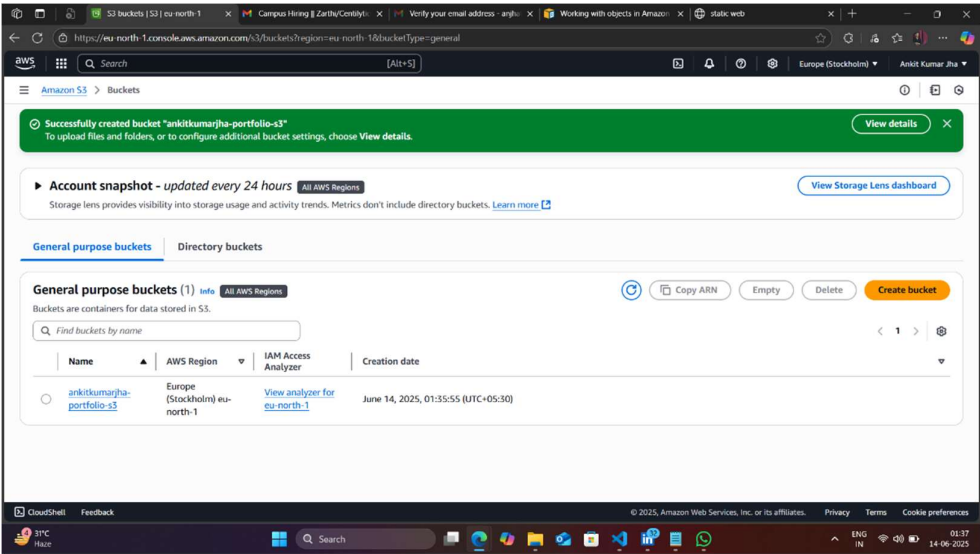
Note: This website contains **http** which is not considered a safe search like **“https”** does.

It may not able to load to some devices like smartphones because this is not considered as safe search and blocked by most mobile browsers for security. You can access this link on computer or by scanning this link through pdf.

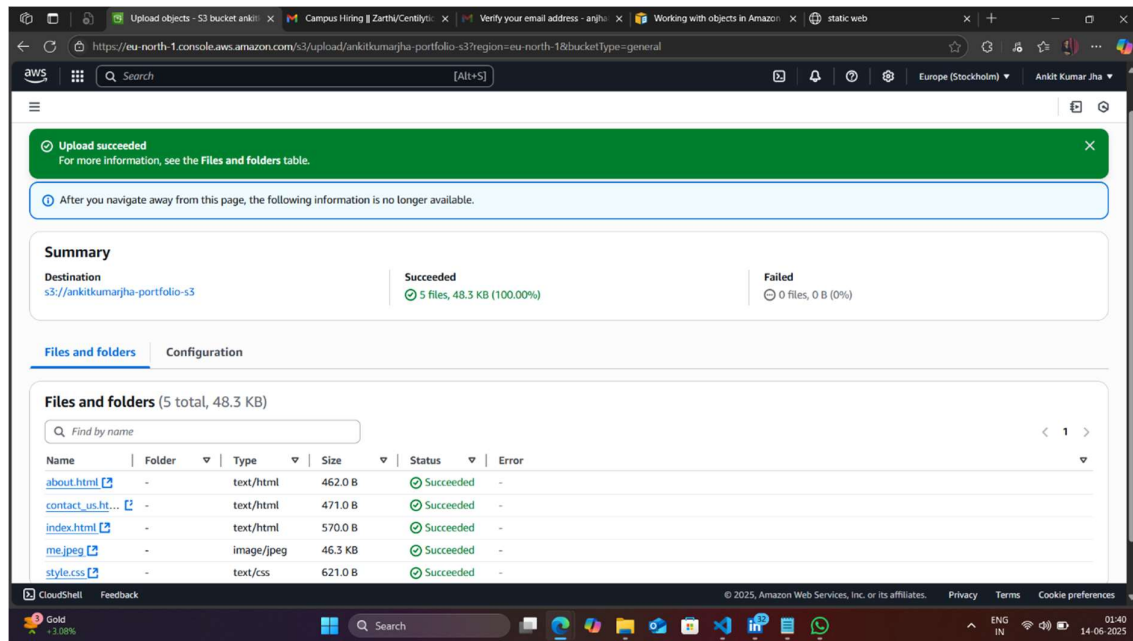
Screenshots



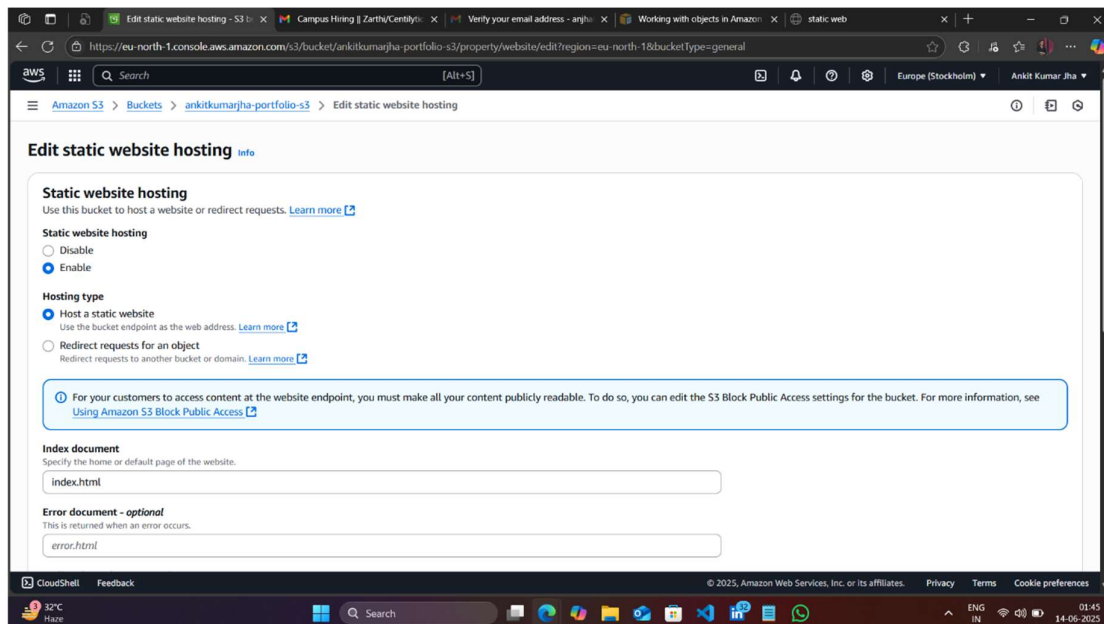
Img 1: Signed up in the AWS account and the page shows AWS console.



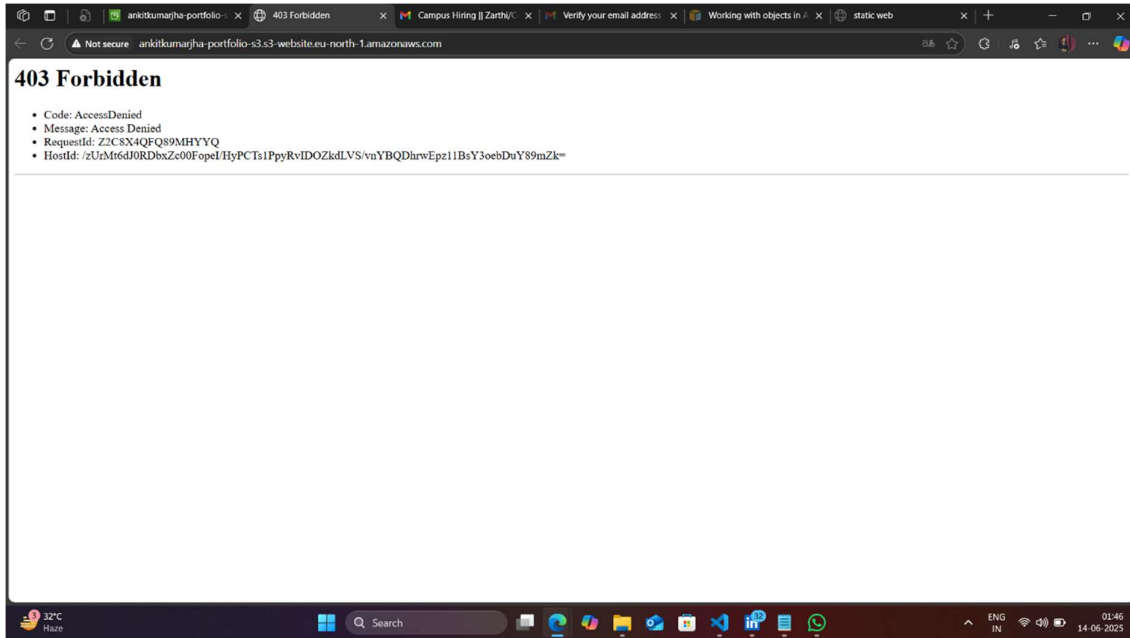
Img 2: Successfully created a first bucket “ankitkumarjha-portfolio-s3”.



Img 3: Here are the files which I uploaded in my bucket.



Img 4: Enabling static website hosting to get links.



Img 5: Error showing 403 Forbidden.

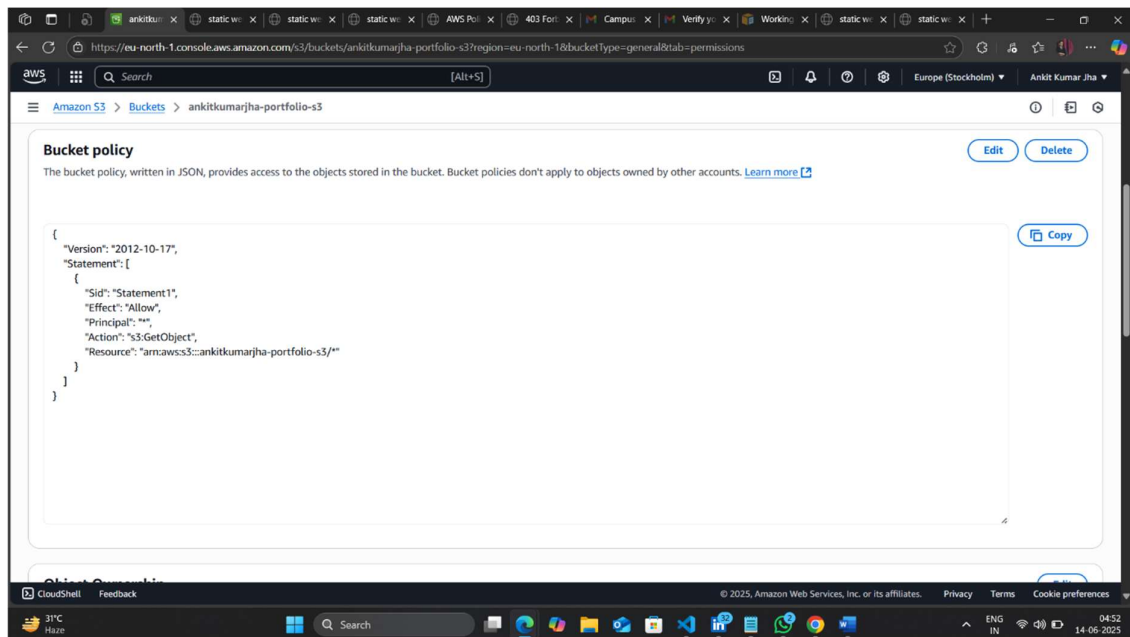


Fig 6: Setting up the bucket policy to allow user read and write.

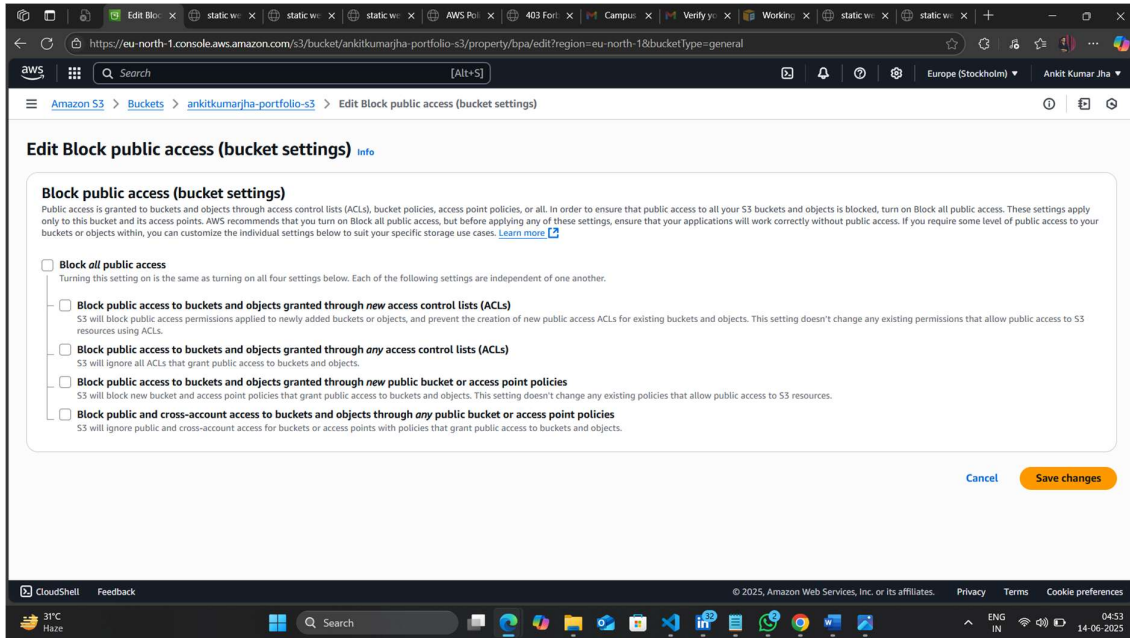


Fig 7: Turning off the block public access for user can access the link.

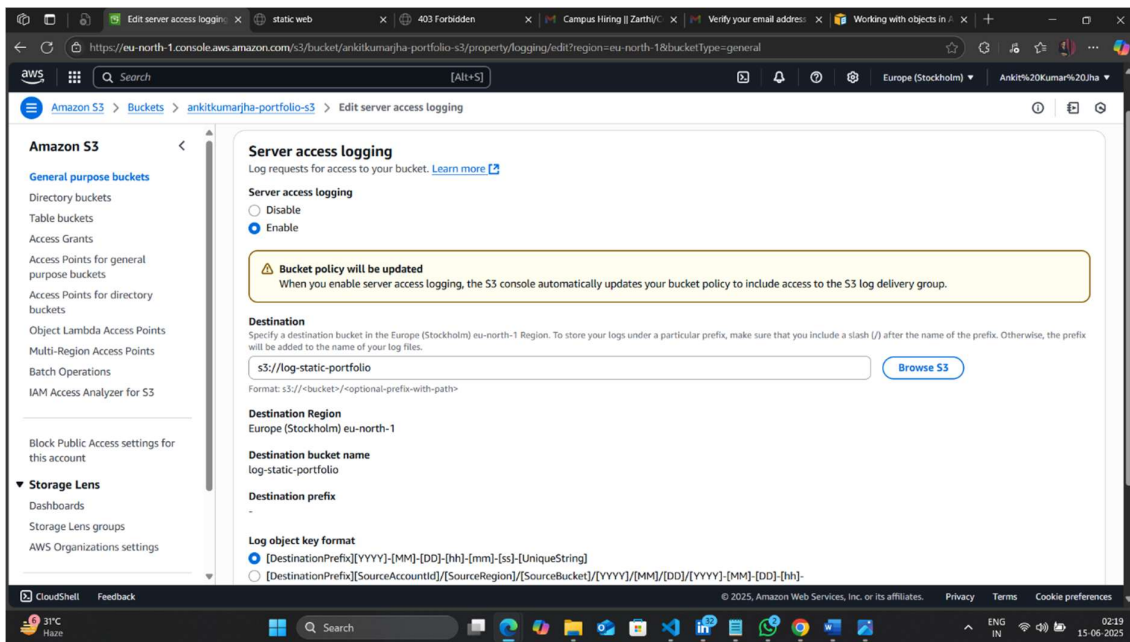


Fig 8: Enabling the server access logging which trace every request made to the bucket.

Issue Summary:

- **Problem:** After successfully uploading my website files to the AWS S3 bucket, I kept getting a 403 error when trying to access the website.
 - ✓ **Sol:** I referred to youtube videos and also read the AWS documentation. I navigated to the S3 bucket permissions and turned off the “Block all public access” setting. I also ensured the bucket policy allowed public read access to all objects. After saving these changes, the website loaded successfully in the browser.
- **Problem:** I didn’t see any updates in the **server access logging** designation to new log bucket.
 - ✓ **Sol:** I went through the official AWS documentation, which helped me to understand that waiting for a few minutes updates the specific bucket and the log appears. I learned that logging in AWS S3 may take some time to reflect updates.
- **Problem:** The website link worked on laptops but did not open on mobile phones.
 - ✓ **Sol:** This website contains “**HTTP**” which is not considered a safe search like “**HTTPS**” does. It may not able to load to some devices like smartphones because this is not considered as safe search and blocked by most mobile browsers for security. We can access this link on computer or by scanning this link through pdf.

Reflection:

Through this assignment, I got to learn that how a cloud computing works, why it plays a crucial role in the world of data storage and server.

As I was a beginner, this assignment was my first hands-on experience with cloud computing, and it gave me valuable insights into:

- How AWS services work especially s3 Static Website Hosting.
- I learned how to create and configure an S3 bucket, manage permissions, set up static hosting, and troubleshoot common issues like access restrictions and error 403.
- I gained the concept of objects in AWS-s3, how a files in buckets is considered as objects.
- Internally objects in AWS-s3 is stored as **Key-Value pair**, where Keys are the unique name and path of the objects and the Value denotes the content inside the Objects.

Overall, I enjoyed the process and realized how cloud platforms like AWS simplify hosting and scaling web content.